United States Patent	12403872
Kind Code	B2
Date of Patent	September 02, 2025
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Hydraulic block for a brake unit of a hydraulic power brake system

Abstract

A bore arrangement of a hydraulic block of a brake unit of a hydraulic power vehicle brake system.

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Appl. No.:	**18/550434**
Filed (or PCT Filed):	**April 12, 2022**
PCT No.:	**PCT/EP2022/059685**
PCT Pub. No.:	**WO2022/263041**
PCT Pub. Date:	December 22, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20240157925 A1	May. 16, 2024

Foreign Application Priority Data

DE	10 2021 206 074.9	Jun. 15, 2021

Publication Classification

Int. Cl.: **B60T7/04** (20060101); **B60T8/36** (20060101); **F15B13/08** (20060101)

U.S. Cl.:

CPC **B60T7/042** (20130101); B60T8/368 (20130101); F15B13/0814 (20130101)

Field of Classification Search

CPC: B60T (8/368); B60T (7/042); B60T (13/686); B60T (13/745); B60T (17/00); F15B (13/0814)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
6688707	12/2003	Dinkel	303/DIG.10	B60T 8/368
8020946	12/2010	Iyatani	303/119.2	B60T 8/368
8523295	12/2012	Bareiss	303/DIG.10	F15B 13/0814
9404514	12/2015	Weh	N/A	F15B 13/0803
9470247	12/2015	Alaze	N/A	F15B 13/0814
9688256	12/2016	Weh	N/A	B60T 8/368
10308232	12/2018	Mayr	N/A	B60T 8/368
11052888	12/2020	Mayr	N/A	B60T 8/368
11485336	12/2021	Ahn	N/A	B60T 13/168
11807200	12/2022	Ahn	N/A	B60T 13/141
11814022	12/2022	Seo	N/A	B60T 13/142
2012/0177516	12/2011	Mayr	417/437	F04B 53/22
2019/0031164	12/2018	Tandler	N/A	B60T 8/368
2019/0210576	12/2018	Zander	N/A	B60T 17/04
2020/0017089	12/2019	Zander	N/A	F15B 13/0842
2020/0094800	12/2019	Zander	N/A	B60T 8/4836
2020/0282968	12/2019	Mayr	N/A	B60T 7/042

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
102006033764	12/2006	DE	B60T 17/02
102012213216	12/2012	DE	N/A
102014207545	12/2014	DE	B60T 17/02
102014213732	12/2015	DE	B60T 13/146
102016202113	12/2016	DE	N/A
102019203308	12/2019	DE	B60T 13/142
2883766	12/2014	EP	N/A
20190090637	12/2018	KR	N/A

OTHER PUBLICATIONS

International Search Report for PCT/EP2022/059685, Issued Jul. 6, 2022. cited by applicant

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Background/Summary

FIELD

(1) The present invention relates to a hydraulic block for a brake unit of a hydraulic power brake system for an automobile.

BACKGROUND INFORMATION

(2) German Patent Application No. DE 10 2016 202 113 A1 describes a narrow, cuboidal hydraulic block for a hydraulic unit of a slip-controlled, hydraulic power brake system having three connections for a brake fluid reservoir in a top side of the hydraulic block. Below the connections, a master brake cylinder bore passes continuously through the hydraulic block, parallel to the top side, from one narrow side to an opposing narrow side adjoining the top side. Below the master brake cylinder bore, a power cylinder bore passes through the hydraulic block transversely to the master brake cylinder bore from one large side of the hydraulic block to an opposing large side of the hydraulic block, which large sides adjoin the top side and the two narrow sides of the hydraulic block. To generate a brake pressure using external power, a power piston can be displaced in the power cylinder bore by an electric motor via a ball screw. The electric motor is arranged externally on the hydraulic block, coaxially to the power cylinder bore, and the ball screw is located between the electric motor and the power piston—likewise coaxially to the electric motor and to the power cylinder bore. The electric motor and the ball screw form a power drive and—together with the power piston and the power cylinder bore—a power brake pressure generator for the hydraulic vehicle brake system. Connections for hydraulic wheel brakes via brake lines are incorporated in one of the two large sides of the hydraulic block in close proximity to the top side at the height of the connections for the brake fluid reservoir. A simulator cylinder bore for a pedal travel simulator of the power brake system is incorporated in a bottom side, opposite the top side, of the conventional hydraulic block.

SUMMARY

(3) A hydraulic block according to the present invention is provided for a brake unit of a hydraulic power brake system for an automobile, which power brake system comprises a brake pressure control system. Brake pressure control refers to the generation and control of a brake pressure in the vehicle brake system, in brake circuits of the vehicle brake system and/or in hydraulic wheel brakes of the brake system, which are connected to the hydraulic block. The brake pressure control system may, in particular, also comprise a slip control system. Slip control systems are, for example, anti-lock brake systems, traction control systems and/or vehicle dynamics management for which the abbreviations ABS, TCS and/or VDM are commonly used. Slip control systems are conventional and are not explained here.

(4) The hydraulic block serves for the mechanical fastening and hydraulic interconnection of hydraulic elements of the vehicle brake system, for the generation of brake pressure and/or for brake pressure control and/or slip control. Such hydraulic elements are, inter alia, solenoid valves, non-return valves, hydraulic accumulators, damper chambers and pressure sensors. The hydraulic elements are fastened in receptacles in the hydraulic block, which are usually formed as cylindrical depressions, blind holes or through holes, in some cases with stepped diameters. “Interconnection” means that the receptacles or the hydraulic elements fastened therein are connected via lines in the hydraulic block according to a hydraulic circuit diagram of the vehicle brake system. The lines are typically, but not necessarily, drilled in the hydraulic block.

(5) Equipped with the hydraulic elements of the vehicle brake system or its slip control system, the hydraulic block forms the brake unit, “equipped” meaning that the hydraulic elements are fastened

in the respective receptacles of the hydraulic block which are provided for them.
(6) The present invention is directed in particular to a bore arrangement of the hydraulic block, i.e. to the routing of lines between the hydraulic elements or their receptacles in the hydraulic block.
(7) The hydraulic block according to an example embodiment of the present invention has a top side, which is provided for seating a brake fluid reservoir. In the top side, the hydraulic block has one or more connections for the brake fluid reservoir.
(8) A fastening side of the hydraulic block, which adjoins the top side of the hydraulic block, is designed for fastening the hydraulic block or the brake unit, i.e., the hydraulic block equipped with the hydraulic elements of the vehicle brake system, to a bulkhead of an automobile. To this end, the hydraulic block has, on the fastening side, for example two standardized internally threaded holes, in which studs or stay bolts can be screwed to fasten the hydraulic block or the brake unit to the bulkhead of the automobile. A master brake cylinder bore leads into the fastening side of the hydraulic block so that a master brake cylinder piston can be displaced in the master brake cylinder bore using muscular power via a brake pedal incorporated opposite the hydraulic block on the bulkhead of the automobile and via a pedal rod, which connects the brake pedal to the master brake cylinder piston in an articulated manner. The master brake cylinder bore preferably extends parallel to the top side in the hydraulic block.
(9) According to an example embodiment of the present invention, a power cylinder bore for generating a brake pressure using external power is incorporated in the hydraulic block transversely to the master brake cylinder bore, between the top side of the hydraulic block and the master brake cylinder bore. The master brake cylinder bore is therefore located below the power cylinder bore, i.e., the master brake cylinder bore is located on a side of the power cylinder bore which is remote from the top side of the hydraulic block or between the power cylinder bore and a bottom side, opposite the top side, of the hydraulic block. The power cylinder bore leads into a motor side of the hydraulic block, which adjoins the top side and the fastening side and is provided for fastening an electric motor for driving a power brake pressure generator.
(10) According to an example embodiment of the present invention, a simulator cylinder bore for a pedal travel simulator is likewise incorporated in the hydraulic block transversely to the master brake cylinder bore, between the top side of the hydraulic block and the master brake cylinder bore, i.e. above the master brake cylinder bore. It preferably leads into a valve side, opposite the motor side, of the hydraulic block and, like the motor side, adjoins the top side and the fastening side.
(11) According to an example embodiment of the present invention, a second return line in the hydraulic block runs from a receptacle for an outlet valve of the vehicle brake system to one of the connections for the brake fluid container through the simulator cylinder bore. The return line is a line for brake fluid, which is produced in the hydraulic block by drilling or in another manner. The second return line preferably runs from the receptacle for the outlet valve to the simulator cylinder bore and, obliquely with respect to the motor side, from the simulator cylinder bore to the connection for the brake fluid container in the top side of the hydraulic block. Moreover, the second return line may run to the master brake cylinder.
(12) According to an example embodiment of the present invention, the second return line is arranged in the hydraulic block such that it leads into the simulator cylinder bore on a rear side of a simulator piston. The simulator piston divides the simulator cylinder bore into a pressurized working chamber on a front side of the simulator piston, which is connected to the master brake cylinder by a simulator valve, and into an unpressurized chamber on the rear side of the simulator piston, which communicates with the unpressurized brake fluid container. According to the present invention, the second return line runs through the unpressurized chamber of the simulator cylinder bore.
(13) According to a configuration of an example embodiment of the present invention, a first return line in the hydraulic block runs from a receptacle for an outlet valve to the power cylinder bore and from the power cylinder bore to one of the connections for the brake fluid container. The two return

lines preferably connect different receptacles for outlet valves to different connections for the brake fluid container. The first return line can be implemented independently of the above-described routing of the second return line.
(14) Through holes or blind holes in the hydraulic block which are referred to as “lines” or “bores” or as “cylinder bores” here may also be produced in a manner other than drilling.
(15) Advantageous configurations and example embodiments of the present invention are disclosed herein.
(16) All of the features disclosed in the description herein and in the figures may be implemented alone or in essentially any combination in specific embodiments of the present invention. Embodiments of the present invention which include only one or multiple features of a specific embodiment of the present invention, rather than all features, are essentially possible. For example, embodiments of the present invention in which the connections for the auxiliary brake unit are arranged at a different point to that described herein are also possible.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The present invention is explained in more detail below with reference to a specific example embodiment depicted in the figures.
(2) FIG. **1** shows a hydraulic circuit diagram of an electrohydraulic power brake system, according to an example embodiment of the present invention.
(3) FIG. **2** shows a motor side of a hydraulic block according to an example embodiment of the present invention for a brake unit of the vehicle brake system of FIG. **1**.
(4) FIG. **3** shows a valve side, opposite the motor side, of the hydraulic block of FIG. **2**, according to an example embodiment of the present invention.
(5) FIG. **3**A shows a section of the hydraulic block along the angled section line A-A in FIG. **3**.
(6) FIG. **3**B shows a section of the hydraulic block along line B-B in FIG. **3**.
(7) FIG. **4** shows part of a bore arrangement of the hydraulic block of FIGS. **2** and **3**.
(8) The figures are partially simplified depictions which are drawn to different scales in some cases.
DETAILED DESCRIPTION OF EXAMPLE EMBODIMENTS
(9) The electrohydraulic power brake system **1** depicted in FIG. **1** is provided for automobiles having four wheel brakes **2**. It has a brake unit **3** to which the four wheel brakes **2** are connected via brake lines. The vehicle brake system **1** is designed as a dual-circuit brake system; two wheel brakes **2** are connected to one brake circuit in each case.
(10) For power braking, the power brake system **3** has a piston-cylinder unit **4** whereof the piston **5** can be displaced in a cylinder **8** by an electric motor **6** via a screw drive **7** as a rotation/translation converter gear. The electric motor **6**, the screw drive **7** and the piston-cylinder unit **4** form a power brake pressure generator **9** of the brake unit **3** for generating a brake pressure for braking using external power. Power braking, for which a brake pressure is generated by the power brake pressure generator **9**, is the normal and intended brake actuation, i.e. service braking.
(11) The power brake pressure generator **9**, i.e. the cylinder **8** of the piston-cylinder unit **4** of the power brake pressure generator **9**, is connected to a chamber **11**′″ of a brake fluid container **11** by a non-return valve **10** and the wheel brakes **2** are connected to the cylinder **8** of the piston-cylinder unit **4** of the power brake pressure generator **9** via a service brake valve **12** in each brake circuit, which service brake valves are hydraulically connected to each other in parallel.
(12) The brake system **1** has, for each wheel brake **2**, an inlet valve **13** and an outlet valve **14**, with which wheel brake pressures in each wheel brake **2** can be controlled individually. The wheel brake pressures in the wheel brakes **2**, and therefore brake forces of the wheel brakes **2**, can thus be

controlled without slip during normal operation. Moreover, slip control systems, such as anti-lock and traction control systems and vehicle dynamics management—which are also referred to colloquially as anti-skid systems—automatic brake systems, distance control systems and the like, are also possible. Such control systems are conventional and are not explained in more detail here. The inlet valves **13** and the outlet valves **14** may also be referred to as wheel brake pressure control valve arrangements **13**, **14**. In each brake circuit, two inlet valves **13** are hydraulically connected to the power brake pressure generator **9** in parallel via a service brake valve **12**.
(13) Besides the power brake pressure generator **9**, the brake system **1** has a dual-circuit master brake cylinder **15** which can be actuated using muscular power and to which the wheel brakes **2** in each brake circuit are connected via an isolation valve **16** and two inlet valves **13**. The master brake cylinder **15** serves as a setpoint value indicator for the wheel brake pressures to be set in the wheel brakes **2**. The brake pressure during power braking is generated by the power brake pressure generator **9**. The master brake cylinder **14** is hydraulically isolated from the wheel brakes **2** during power braking by closing the isolation valves **16**. In the event of a failure of the power brake pressure generator **9**, the brake pressure is generated by actuating the master brake cylinder **15** using muscular power (so-called auxiliary braking).
(14) So that, when the isolation valves **16** are closed, brake fluid may be displaced from the master brake cylinder **15** and pistons of the master brake cylinder **15** and a brake pedal **17** may be moved, the brake unit **3** has a pedal travel simulator **18**, which is connected to the master brake cylinder **15** in a brake circuit via a simulator valve **19**. The pedal travel simulator **18** is a piston-cylinder unit having a spring-loaded piston.
(15) In the described and depicted specific embodiment of the present invention, the inlet valves **13** and the isolation valves **17** are 2/2-way solenoid valves which are open in their currentless normal positions and the service brake valves **12** of the power brake pressure generator **9**, the outlet valves **14** and the simulator valve **19** are 2/2-way solenoid valves which are closed in their currentless normal positions.
(16) The hydraulic elements of the electrohydraulic power brake system **1**, namely the valves **12**, **13**, **14**, **16**, **19**, **21** of the power brake pressure generator **9**, the master brake cylinder **15**, the pedal travel simulator **18** and further elements such as pressure sensors, are arranged in receptacles of a hydraulic block **20** of the brake unit **3** and connected to each other via a bore arrangement of the hydraulic block **20** according to the hydraulic circuit diagram (depicted in FIG. **1**) of the vehicle brake system **1**. A receptacle **21′** for the test valve **21** communicates with the master brake cylinder bore **15′** via a bore **42** which runs axially from a base of the receptacle **21′** to a groove **38** surrounding the master brake cylinder bore **15′** (FIG. **3**B). The groove **38** and the bore **42** leading into it are shown offset into the section plane in FIG. **3**B. They are actually located behind the bore **42** and the oblique bore **39** as viewed looking onto the fastening side **29** of the hydraulic block **20**.
(17) Moreover, an oblique bore **39** runs from the base of the receptacle **21′** for the test valve **21** to a horizontal bore **40**, which runs to the first return line **22** between the master brake cylinder bore **15′** and the valve side such that it is parallel to the master brake cylinder bore **15′**, whereby the receptacle **21′** for the test valve **21** is connected to one of the connections **27′** for the brake fluid container **11** (FIG. **3**B).
(18) In one of the two brake circuits, the master brake cylinder **15** is connected to one of the chambers **11′** of the brake fluid reservoir **11** via a test valve **21**. In the exemplary embodiment, the test valve **21** is likewise a 2/2-way solenoid valve which is open in its currentless normal position. In the other brake circuit, the master brake cylinder **15** is connected to a rear side of the pedal travel simulator **19**, which is in turn connected to a chamber **11″** of the brake fluid container **11** and is therefore unpressurized. The rear side of the pedal travel simulator **19** refers to one of two chambers of the cylinder of the pedal travel simulator **19** into which the cylinder of the pedal travel simulator **19** is divided by the piston of the pedal travel simulator **19**. A front side of the pedal travel simulator **19** is connected to the master brake cylinder **15** via the simulator valve **19** and is

pressurized by the master brake cylinder **15** when the simulator valve **19** is open.

(19) In one brake circuit, the outlet valves **14** are connected to a chamber **11′** of the brake fluid container **11** via a first return line **22**, which, in the cylinder **8** of the piston-cylinder unit **4** of the power brake pressure generator **9**, runs around the piston **5** of the piston-cylinder unit **4**. In the other brake circuit, the outlet valves **14** are connected to another chamber **11″** of the brake fluid container **11** via a second return line **23** together with the master brake cylinder **15**.

(20) FIG. **2** shows a motor side **24** and FIG. **3** shows a valve side **25** of the hydraulic block **20** of the brake unit **3**. In the exemplary embodiment, the hydraulic block **20** is a flat, cuboidal metal block, which serves for the mechanical fastening and hydraulic interconnection of the hydraulic elements of the power brake system **1**. Equipped with the hydraulic elements, the hydraulic block **20** forms the brake unit **3** of the vehicle brake system **1**. “Flat” means that the hydraulic block **20** is approximately three to four times as wide or long as it is thick. In the exemplary embodiment, two opposing large sides of the hydraulic block **20** are virtually cuboidal and form the motor side **24** and the valve side **25**. In FIGS. **2** and **3**, the hydraulic block **20** is shown in the unequipped state, i.e. without the hydraulic elements.

(21) A narrow side of the hydraulic block **20**, which is referred to as the top side **26** here, has three cylindrical blind holes as connections **27′**, **27″**, **27‴** for the chambers **11′**, **11″**, **11‴** of the brake fluid container **11**, which is arranged on the top side **26** of the hydraulic block **20** (not depicted). At the bottom of the brake fluid reservoir **11**, connection nipples reach into the blind holes of the hydraulic block **20**, which form the connections **27′**, **27″**, **27‴**, and are sealed there using O rings.

(22) In the hydraulic block **20**, a master brake cylinder bore **15′**, which forms the master brake cylinder **15**, is incorporated parallel to the top side **26** and in a center between the motor side **24** and the valve side **25**. In FIGS. **2** and **3**, the master brake cylinder bore **15′** is depicted by dashed lines. In the exemplary embodiment, it is located somewhat below a center of the hydraulic block **20** between the top side **26** and an opposing bottom side **28** of the hydraulic block **20**, so that the master brake cylinder bore **15′** lies approximately tangentially against a center plane of the hydraulic block **20** between the top side **26** and the bottom side **28**.

(23) The master brake cylinder bore **15′** is open at a narrow side of the hydraulic block **20**; it leads into this narrow side of the hydraulic block **20** or has a mouth there. The narrow side of the hydraulic block **20**, in which the master brake cylinder bore **15′** is open or into which it leads, is referred to here as the fastening side **29** of the hydraulic block **20**. It adjoins the top side **26**, the motor side **24**, the valve side **25** and the bottom side **28** of the hydraulic block **20** and is provided for fastening the hydraulic block **20** to a bulkhead (not depicted) of the automobile. The hydraulic block **20** is fastened to the bulkhead of the automobile such that the top side **26** with the brake fluid container **11** is located at the top. The master brake cylinder bore **15′** is open at the fastening side **29** of the hydraulic block **20** so that a master brake cylinder piston can be displaced in the master brake cylinder bore **15′** by the foot brake pedal **17**, which is fastened to an opposing side of the bulkhead, via a pedal rod which connects the master brake cylinder piston to the foot brake pedal **17** in an articulated manner. The foot brake pedal **17** and the pedal rod are not shown in FIGS. **2** and **3**.

(24) A power cylinder bore **8′**, which forms the cylinder **8** of the power brake pressure generator **9**, is incorporated perpendicularly to the master brake cylinder bore **15′** in the motor side **24** of the hydraulic block **20** and protrudes over the valve side **25** in the manner of a dome **30**. The power cylinder bore **8′** is located somewhat above the master brake cylinder bore **15′**, i.e. between the master brake cylinder bore **15′** and the top side **26** of the hydraulic block **20**. The power cylinder bore **8′** passes the master brake cylinder bore **15′** orthogonally with a slight spacing. It is arranged somewhat eccentrically offset in the direction of the fastening side **29** of the hydraulic block **20**.

(25) The electric motor **6** of the power brake pressure generator **9**, which is not shown in FIG. **2**, is arranged externally on the motor side **24** of the hydraulic block **20**, coaxially to the power cylinder bore **8′**. A planetary gear as a reducing gear and the screw drive **7**, which is a ball screw in the

exemplary embodiment, are arranged coaxially to the power cylinder bore **8′** between the electric motor **6** and the piston **5** of the power brake pressure generator **9** (not shown in FIG. **2**).
(26) A cylinder bore **18′** of the pedal travel simulator **18** is incorporated in the valve side **25** of the hydraulic block **20** such that it is parallel to the power cylinder bore **8′** and perpendicular to the master brake cylinder bore **15′**. In the exemplary embodiment, the cylinder bore **18′** is located between the master brake cylinder bore **15′** and the top side **26** of the hydraulic block **20** and in closer proximity to the top side **26** than to the master brake cylinder bore **15′** and between the power cylinder bore **8′** and a narrow side **31**, opposite the fastening side **29**, of the hydraulic block **20**.
(27) Receptacles **12′**, **13′**, **14′**, **16′**, **19′**, **21′** for the solenoid valves **12**, **13**, **14**, **16**, **19**, **21** and receptacles for further elements, such as pressure sensors, are incorporated in the valve side **25** of the hydraulic block **20**. The receptacles, which, in FIG. **3**, are denoted by the reference numeral of the respective solenoid valve **12**, **13**, **14**, **16**, **19**, **21** or other element followed by "′" are cylindrical depressions or blind holes in the hydraulic block **20**, in some cases with stepped diameters. The hydraulic elements are inserted into the receptacles and caulked around the circumference in a pressure-tight manner. Hydraulic portions of the solenoid valves **12**, **13**, **14**, **16**, **19**, **21**, which form the actual valves, are located in the receptacles; armatures and magnetic coils, which are accommodated in a valve dome, protrude from the valve side **25** of the hydraulic block **20**.
(28) The hydraulic block **20** of the brake unit **3** is bored according to the hydraulic circuit diagram shown in FIG. **1**. "Bored" or "bore arrangement" refer to the cylinder bores, receptacles for the solenoid valves and connections which are incorporated in the hydraulic block **20** as well as the bores forming the lines connecting them according to the hydraulic circuit diagram. The hydraulic block **20** is Cartesian-bored, i.e. the bores, receptacles, connections, lines etc. are incorporated in the hydraulic block **20** such that they are parallel and perpendicular to each other and to sides and edges of the hydraulic block **20**. This does not exclude individual, obliquely extending lines and bores.
(29) Two connections **2′** for brake lines which run to two wheel brakes **2** are incorporated in the motor side **24** along the narrow side **31**, opposite the fastening side **29**, and two connections **2′** for brake lines which run to two wheel brakes **2** are incorporated in the motor side **24** in close proximity to the bottom side **28** of the hydraulic block **20**. "In close proximity" means a spacing of no more than a radius of the respective connection.
(30) Between the power cylinder bore **8′** and the top side **26**, three continuous through holes from the motor side **24** to the valve side **25** are incorporated in the hydraulic block **20** as motor connection bores **32** for supplying power to the electric motor **6** of the power brake pressure generator **9**. The motor connection bores **32** are incorporated in the hydraulic block **20** on an imaginary arc around the power cylinder bore **8′**, between the power cylinder bore **8′** and the top side **26**. A signal bore **33** for control lines and/or signal lines to or from the electric motor **6** is likewise incorporated in the hydraulic block **20** on the imaginary arc on which the motor connection bores **32** are incorporated.
(31) The first return line **22**, which is formed as a bore in the hydraulic block **20** and connects two of the receptacles **14′** for the outlet valves **14** of two wheel brakes **2** to one of the connections **27′** for one of the chambers **11′** of the brake fluid container **11** around the piston **5** of the piston-cylinder unit **4** of the power brake pressure generator **9**, is depicted by dashed lines in FIG. **3**. Starting from the connection **27′** for the chamber **11′** of the brake fluid container **11**, the first return line **22** runs firstly as an oblique bore from a base of the connection **27′** to the power cylinder bore **8′**, which forms the cylinder **8** of the piston-cylinder unit **4** of the power brake pressure generator **9**. From the power cylinder bore **8′**, the first return line **2** extends further downwards, with a parallel offset in the direction of the fastening side **29**, between the two receptacles **14′** for the two outlet valves **14** through to the bottom side **28** of the hydraulic block **20**. The two receptacles **14′** are connected to the first return line **22** via transverse bores, which lead into the fastening side **29** of

the hydraulic block **20** (not depicted). "Oblique bore" means that the portion of the first return line **22** from the connection **27′** for the brake fluid reservoir **11** runs to the power cylinder bore **8′** obliquely with respect to the motor side **24** and with respect to the valve side **25** and—in the exemplary embodiment—parallel to the fastening side **29** and to the narrow side **31**. Except for the portion from the connection **27′** to the power cylinder bore **8′**, the first return line **22**, like all other bores, extends parallel to the sides **24**, **25**, **26**, **28**, **29**, **31** and edges of the hydraulic block **20**. The portion of the first return line **22** which comes from the connection **27′** for the brake fluid container **11**, and likewise the portion of the first return line **22** which runs to the receptacles **14′** for the outlet valves **14**, lead into a groove **34** in the power cylinder bore **8′**, which is located in a radial plane of the power cylinder bore **8′** in which the piston **5** of the piston-cylinder unit **4** of the power brake pressure generator **9** is located when it is installed. The first return line **22** thus runs around the piston **5** of the piston-cylinder unit **4** of the power brake pressure generator **9** without communicating with the cylinder **8** of the piston-cylinder unit **4** of the power brake pressure generator **9**. The groove **34** in the power cylinder bore **8′** is denoted by dashed lines in FIG. **3**.
(32) The second return line **23** is likewise depicted by dashed lines in FIG. **3**. It runs upwards from the two other receptacles **14′** for the outlet valves **14** of two wheel brakes **2** to the cylinder bore **18′** of the pedal travel simulator **18** and, from there, further upwards, with a parallel offset in the direction of the narrow side **31** of the hydraulic block **20**, to another connection **27″** of a chamber **11″** of the brake fluid container **11** as the first return line **22**.
(33) Between the cylinder bore **18′** of the pedal travel simulator **18** and the connection **27″** of the chamber **11″** of the brake fluid container **11**, the second return line **23** extends parallel to the fastening side **28** and to the narrow side **31**, opposite this fastening side, of the hydraulic block **20**, albeit obliquely with respect to the motor side **24** and with respect to the valve side **25** of the hydraulic block **20**. The oblique extent of the second return line **23** can be seen in FIG. **3**A. The section plane of FIG. **3**A has a radial offset with respect to the cylinder bore **18′** of the pedal travel simulator **18** in the direction of the fastening side **29** or in the direction of the narrow side **31**, opposite the fastening side **29**, of the hydraulic block **20**. The offset of the section plane of FIG. **3**A is shown by the section line A-A in FIG. **3**. The portion of the second return line **23** which extends obliquely with respect to the motor side **24** and with respect to the valve side **25** of the hydraulic block **20** between the cylinder bore **18′** of the pedal travel simulator **18** and the connection **27″** of the chamber **11″** of the brake fluid container **11** is shown offset into the section plane in FIG. **3**A. The oblique portion of the second return line **23** actually has a radial offset with respect to the connection **27″** of the chamber **11″** of the brake fluid container **11**, which can be seen in FIG. **3**.
(34) A transverse bore **37** connects the two receptacles **14′** for the outlet valves **14** to each other.
(35) The second return line **23** leads into the master brake cylinder bore **15′** so that one of the chambers of the dual-circuit master brake cylinder **15** is connected to one of the chambers **11″** of the brake fluid container **11** via the cylinder bore **18′** of the pedal travel simulator **18**. The second return line **23** leads into a radial plane of the cylinder bore **18′** of the pedal travel simulator **18**, which is located on a rear side of the piston of the pedal travel simulator **18** and is unpressurized as a result of its connection to the brake fluid container **11**.
(36) The second return line **23** is drilled into the hydraulic block **20** from the bottom side **28** and passes radially through the master brake cylinder bore **15′** so that a drill bit, when drilling the second return line **23**, meets a cylinder wall of the master brake cylinder bore **15′** perpendicularly, and not obliquely, as it exits the master brake cylinder bore **15′**. As a result, as it exits the master brake cylinder bore **15′**, the drill bit is not offset tangentially to the master brake cylinder bore **15′** and does not break off. The master brake cylinder bore **15′** has a circumferential groove **41** into which the second return line **23** leads radially at two diametrically opposed points.
(37) FIG. **4** shows a fragment of the hydraulic block **20** as viewed looking onto the narrow side **31** opposite the fastening side **29**. The dome **30**, which elongates the power cylinder bore **8′** and protrudes from the valve side **25** of the hydraulic block **20**, can be seen. The power cylinder bore

8′, like the connections **27″**, **27‴** of the brake fluid container **11**, is shown by solid lines. A brake line, which is formed as a bore in the hydraulic block **20** and leads coaxially into one of the connections **27‴**, extends—after bending parallel to the power cylinder bore **8′**—in a wall of the dome **30** outside the power cylinder bore **8′** and leads into a circumferential groove **36**, which surrounds the power cylinder bore **8′** in the dome **30**.
(38) The connection **27‴** for the brake fluid container **1** has a depression **10′** elongating it as a receptacle for the non-return valve **10** in the brake line **35** which runs to the power cylinder bore **8′**. The non-return valve **10** is depicted as a circuit symbol in FIG. **4**.
(39) To vent the master brake cylinder **15**, i.e. to enable air bubbles to be removed from the brake fluid in the master brake cylinder **15**, a brake line **43**, which connects the master brake cylinder bore **15′** to one of the receptacles **16′** for the two isolation valves **16**, leads tangentially into the master brake cylinder bore **15′** at a circumferential point which faces the top side **26** of the hydraulic block **20** (FIG. **3**B). On the right in FIG. **3**, in close proximity to the mouth of the power cylinder bore **15′** on the fastening side **29** of the hydraulic block **20**, the tangential arrangement of one of the receptacles **16′** for an isolation valve **16** with respect to the top side **26** of the hydraulic block **20** can be seen.

Claims

1. A hydraulic block for a brake unit of a hydraulic power brake system for an automobile, the hydraulic block comprising: a top side, which is configured for seating a brake fluid container and has connections for the brake fluid container; a fastening side, which adjoins the top side, is configured for fastening the hydraulic block to a bulkhead of the automobile and into which a master brake cylinder bore leads; a motor side, which adjoins the top side and the fastening side, into which a power cylinder bore leads and on which an electric motor can be arranged for displacement of a power piston of a power brake pressure generator in the power cylinder bore; connections for hydraulic wheel brakes; and a simulator cylinder bore for a pedal travel simulator of the power brake system; wherein the power cylinder bore and the simulator cylinder bore are incorporated in the hydraulic block between the top side and the master brake cylinder bore, and a second return line in the hydraulic block runs from a receptacle for an outlet valve to one of the connections for the brake fluid container through the simulator cylinder bore.
2. The hydraulic block as recited in claim 1, wherein the second return line runs from the receptacle for the outlet valve to the master brake cylinder bore and, obliquely with respect to the motor side, from the simulator cylinder bore to the one of the connections for the brake fluid container.
3. The hydraulic block as recited in claim 1, wherein the second return line passes radially through the master brake cylinder bore.
4. The hydraulic block as recited in claim 1, wherein a first return line in the hydraulic block runs from the receptacle for the outlet valve to the power cylinder bore and from the power cylinder bore to the one of the connections for the brake fluid container.
5. The hydraulic block as recited in claim 1, wherein a first portion of the first return line which comes from the receptacle for the outlet valve leads into a groove in the power cylinder bore at a different circumferential point to a second portion of the first return line which runs to the one of the connections for the brake fluid container.
6. The hydraulic block as recited in claim 1, wherein the hydraulic block has a dome, which protrudes from a valve side, opposite the motor side, and into which the power cylinder bore extends, and a portion of a brake line which runs in the hydraulic block from the one of the connections for the brake fluid container in the top side of the hydraulic block to the power cylinder bore, extends through a cylinder wall of the dome.
7. The hydraulic block as recited in claim 1, wherein the hydraulic block has receptacles for valves of a brake pressure control system of the power brake system in a valve side of the hydraulic block,

opposite the motor side, of the hydraulic block.
8. The hydraulic block as recited in claim 1, wherein the connections for the wheel brakes are arranged in the motor side of the hydraulic block.
9. The hydraulic block as recited in claim 1, wherein a brake line, which runs in the hydraulic block from the master brake cylinder bore to a receptacle for an isolation valve, leads into the master brake cylinder bore at a circumferential point which faces the top side of the hydraulic block.
10. The hydraulic block as recited in claim 1, wherein the one of the connections for the brake fluid container in the top side of the hydraulic block has a concentric receptacle for a non-return valve, from which a brake line in the hydraulic block runs to the power cylinder bore.
11. The hydraulic block as recited in claim 1, wherein: i) motor connection bores for supplying power to the electric motor of the power brake pressure generator, and ii) a signal bore for control lines and/or signal lines to or from the electric motor, are incorporated in the hydraulic block such that they pass through the hydraulic block from the motor side to a valve side on an imaginary arc around the power cylinder bore.